Can choosing which radiocarbon dates to keep rewrite the story of the past?

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The puzzle

Archaeologists reconstruct how human activity rose and fell over thousands of years by counting dated traces of people: charcoal from a hearth, shell from a midden. More people leave more traces, so a curve showing how many *radiocarbon dates* (age estimates from the decay of carbon-14 in once-living material) fall in each century is read as a history of human activity. Regional databases now pool thousands of such dates.

A site may have been dated more than once while a database lists only some of its dates. A listed date may be the site's only one, or one kept from several: the oldest, say, because a study asked when people first arrived. Nothing in the date itself records which.

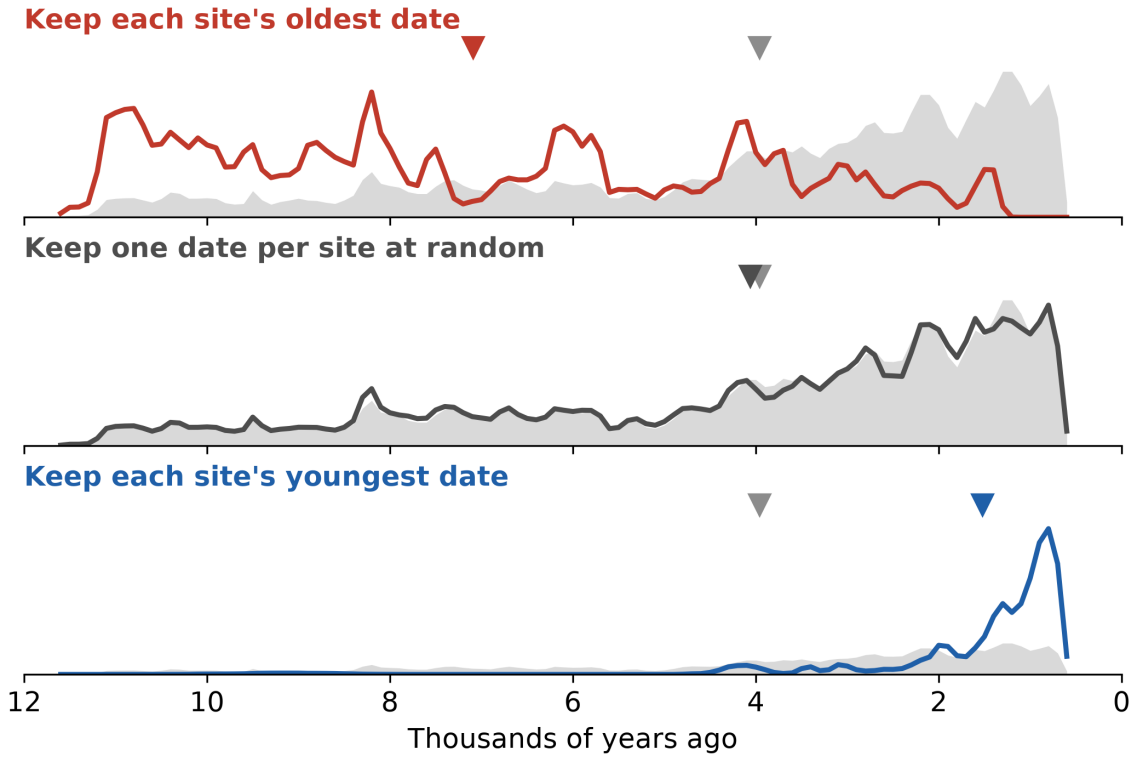


Figure. The same 540 dates from 68 Australian sites after keeping one date per site in three ways. Grey is the curve from all 540 dates; triangles mark each curve's average age.

Why it is hard

Each date has its own measurement uncertainty, and careful methods already handle that. The problem here is different: which dates were kept. An extreme is not a typical member of a set. Of five dates from a steady history, the oldest falls in the oldest fifth of the time span about two times in three, against one time in five for a date picked at random. A curve built from oldest dates therefore tends to be pushed back in time even if nothing in the past changed. Agreement among standard ways of drawing the curve cannot, by itself, reveal this, because every method starts from the same dates.

What we did

- We took 540 well-replicated charcoal and wood dates from 68 Australian archaeological sites and redrew the regional curve after keeping each site's oldest date, its youngest date or a date picked at random.

- We simulated archives whose true history we knew (steady, rising or falling, by tenfold or by as little as one and a half times over 11,000 years) and asked whether standard model comparison recovered the true direction after each kind of selection.
- We repeated the main tenfold simulations with measurement scatter added to every simulated date.

What we found

1. **Keeping each site's oldest or youngest date moved the Australian curve by millennia.** Its average age shifted 3,133 years older or 2,439 years younger, although both versions kept exactly 68 dates. Keeping one date per site at random moved it by about 100 years. Oldest and youngest retention changed the curve's shape more than the choice among three standard ways of drawing it did. They show how far selection can move this collection of dates, not which curve is the true history.
2. **When every site was affected, the selection rule decided the answer.** In the simulations, keeping every site's oldest date always produced an apparent decline, and keeping its youngest always an apparent rise, whatever the true history; true trends running the other way were reversed. Applying the rule to only half the sites still created a trend from a steady history in over 97% of runs.
3. **Weaker true trends were reversed even when only some sites were affected.** With three-quarters of sites affected, gentle trends (a one-and-a-half- to twofold change over 11,000 years) were reversed in 93–100% of runs, although no tenfold trend was. The proportions we tested do not establish a safe threshold.
4. **Keeping both the oldest and youngest dates avoided reversals but blurred the picture.** It never reversed a trend. With every site affected, it detected only 11–17% of the gentlest trends, against 61–67% for two dates per site picked at random, and it always made tenfold straight-line trends look exponential.
5. **Random selection almost never reversed a trend, and measurement scatter did not change the main results.** Picking dates at random reversed at most 0.6% of runs, though with every site affected it often missed gentle trends. Adding measurement scatter to the tenfold simulations moved no headline result by more than 0.1 percentage points; weaker trends were not rerun, and errors shared between dates through the calibration curve, laboratory offsets and outliers remain untested.

What it means

A regional curve inherits the choices made in assembling its dates. Equal numbers of dates per site do not make those dates representative, and agreement among different methods applied to the same dates cannot, by itself, show that they are. Databases should record whether a site's date is the only one obtained or one chosen from several, and on what basis. Analysts should test whether their conclusions survive plausible selection rules. The simulations treat dates as independent draws without site histories, and we do not know how often real archives keep only extreme dates.

How this work was done

Barry Brook set the questions and directs the work; the human co-authors contribute archaeological and statistical expertise and will review this draft. Codex (OpenAI GPT-5.6 and GPT-6) and Claude (Anthropic Fable 5.1 and Opus 5.5) designed and ran the analyses, audited each other's evidence and drafted the text. Every number here traces to an independently checked claims ledger in the repository.

Sources

The article, supplement and an agent guide are in the repository (`manuscript/`). The dates come from the AustArch database (Williams and Ulm, 2014).

Williams, A., Ulm, S., 2014. AustArch: A database of 14C and luminescence ages from archaeological sites in Australia. https://doi.org/10.5284/1027216